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Github
Blog

Jo Hyenogseob

BIO

I am currently a master's student in the Department of Computer Science and Engineering at Dongguk University. I am a member of the Smart Vision & Media (SVM) Lab, focusing on computer vision and deep learning. My research spans both vision inspection for automated manufacturing environments and multimodal learning that integrates visual and textual information to build more robust AI systems.

FIELD OF INTEREST

Computer vision, Representation learning, Defect classification, Multimoal learning

EDUCATION

Dongguk University (DGU) *Expected Feb. 2026*
M.S. in Computer Science and Engineering (CSE)

Dongguk University (DGU) *Feb. 2024*
B.S.E. in Mechanical Engineering (ME)

Gyeongkuk National University (GKNU) *Feb. 2020*
B.A. in Ethics Eduaction, Double Major English Education

RESEARCH EXPERIENCE

Multi-Illumination Display Inspection *Mar. 2024 - Feb. 2025*
Graduate Student Research Assistant *LGDisplay*

- Developed a defect classification algorithm for display panels using multi-illumination image inputs.
- Conducted in collaboration with LG Display as part of an industry-academia joint research project.
- Achieved 98.5% classification accuracy on defect test images and deployed the algorithm within LG Display's production inspection system.

POSITIONS

Graduate Student Research Assistant (RA) *Mar. 2024 - Jun. 2025*
Smart Vision & Media (SVM) Lab
Computer Science and Engineering (CSE), Dongguk University (DGU)

Contract Teacher *Mar. 2020 - Dec. 2020*
Social Studies, Ethics
Youngju Girls' High School

PROJECT

2025 Samsung Collegiate Programming Challenge: AI challenge Summer 2025
AI Competition Project: Multimodal AI for Everyday Photo Understanding *Samsung Research*

- Proposed CLIP · SigLIP-based ensemble model for multimodal image-text classification.
- Achieved 71% accuracy on test set (851 samples), ranked 41st out of 242 participants (Top 50 finalist).

LG Aimers Winter 2023
AI Competition Project: B2B Sales Opportunity Prediction *LG AI Research*

- Developed a binary classification model to predict B2B sales opportunities from MQL data.
- Benchmarked 10+ ML models (e.g., XGBoost, LightGBM) for B2B sales opportunity prediction.

Autonomous Robot Navigation Spring 2023
Course Project (MEC4092-01) *Dongguk University*

- Implemented ROS2-based autonomous navigation with real-time control and sensor-fusion.
- Evaluated navigation performance in a physical testbed environment with obstacle avoidance tasks.

PUBLICATIONS

International Conferences

1. Hyeongseob Jo, Seunggi Park, and Sung In Cho. A Survey of Unsupervised Learning-Based Out-of-Distribution Detection. In *Proceedings of the IEEE/IEIE International Conference on Consumer Electronics-Asia (ICCE-Asia)*, Nov. 2024
2. Y. S. Oh, M. G. Kim, B. Kim, J. Y. Kim, Hyeongseob Jo, J. H. Park, G. Lee, and S. I. Cho. Dual-Branch Network via Multiple Illumination-Aware Representation Learning for Steel Surface Defect Classification. In *Proceedings of the British Machine Vision Conference (BMVC)*, Sheffield, UK, Nov. 2025
3. Yoonseo Park, Hyeongseob Jo, and Sung In Cho. A Survey of Training-free Diffusion-based Image Generation with Free-form Mask. In *Proceedings of the 2025 International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC)*, Jul. 2025

Domestic Conference

1. 이선호, 남혁, 조형섭, and 조성인. 차선 검출: 주요 접근법과 최신 연구 동향. In *대한전자공학회 하계학술대회 논문집*, pages 2969–2973, 2025

Magazine Article

1. 조형섭, 오용석, and 조성인. 다중 조명 환경에서 제품 외관 검사 기법에 대한 이해. *Information Display (한국정보디스플레이학회)*, Dec. 2025

Dissertations

1. Hyeongseob Jo. 산업 공정의 다중 조명 환경에서 표면 결함 자동 검사를 위한 경량 딥러닝 기반 학습방법. Master's thesis, Dongguk University, Seoul, Korea, Feb. 2026

PATENTS

Domestic Patent

[1] 조성인, 조형섭, 이미지 내 결함을 분류하는 시스템 및 분류 방법, 출원 *10-2025-0155364* (2025-10-24).

TEACHING ASSISTANT

AI course (machine learning, deep learning) *Jun. 10–13 and 23–25 2025*
LG Display

Introduction to Industry-Academia Project (CAI7177-01) *Spring 2025*
Computer Science and Engineering (CSE), Dongguk University (DGU)

Signal Processing, Image Processing *Aug. 19–23 2024*
LX Semicon

Creative Engineering Design (CSC2004-03) *Spring 2024*
Computer Science and Engineering (CSE), Dongguk University (DGU)

LANGUAGES & SKILLS

- Korean (native), English (conversational)
- Pytorch, MATLAB, C
- LaTeX, Microsoft Office, Ubuntu, Windows

Revised December 2, 2025